

# Immune memory: recognise, respond, remember

*Teacher guide and worked answers*

**Explain how barriers and the specific immune response protect the body, including memory after immunisation.**

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

## Scheme of work

\_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx | LAUNCH Weekly - Spring | C34

Explain the body's defences & immune response.

Directly develops Pearson 5.12-5.14; antibiotics are a short 5.16 contrast. Core mechanism is Foundation-accessible. This is a conceptual model, not medical advice.

## Prior learning

A pathogen can cause disease.

Different diseases have different causes.

A line graph shows change using stated axes.

## Success evidence

Connect a barrier with how it reduces entry.

Distinguish an antigen, an antibody and a memory lymphocyte.

Use a graph to explain the faster secondary response to the same antigen.

## Equipment

Printed shared evidence and one selected response route; pen, ruler and calculator.

Projector or offline HTML/PowerPoint. Optional word processor or spreadsheet.

## Preparation

Print the shared evidence page, one chosen response route and exit page. Routes are alternatives, not a workload ladder.

Read the teacher answers before teaching. All numerical sample data in this lesson are constructed for learning.

## Practical care

Paper and screen tasks; no biological sampling, cultures, medicine handling or personal health disclosures.

## Ready to teach alternative

All evidence is included. Point, annotate, dictate or discuss quietly instead of writing or using a device.

# Teaching sequence

| Slide | Stage     | Minutes      | Focus                                  |
|-------|-----------|--------------|----------------------------------------|
| 1     | Opening   | 0-2          | How can the second response be faster? |
| 2     | Retrieval | 2-5          | Three distinctions                     |
| 3     | I do      | 5-9          | Before a specific response             |
| 4     | I do      | 9-12         | Recognise the antigen                  |
| 5     | I do      | 12-15        | Read the time and amount               |
| 6     | We do     | 15-18        | Build the explanation chain            |
| 7     | We do     | 18-20        | Choose a target, test the fit          |
| 8     | We do     | 20-22        | Evidence desk: compare day 6           |
| 9     | Hinge     | 22-25        | Which mechanism is correct?            |
| 10    | You do    | 25-26        | Create a defence explanation           |
| 11    | You do    | 26-36        | Make the model teach someone           |
| 12    | Review    | 36-38        | Audience test: is the mechanism clear? |
| 13    | Review    | 38-39        | Improve the final claim                |
| 14    | Exit      | 39-40        | Explain the faster response            |
| 15    | Exit      | Within stage | Choose the next useful step            |

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

## 1 Opening How can the second response be faster?

Invite prediction. Explain that “memory” is a property of immune cells, not a person remembering to fight an infection.

## 2 Retrieval Three distinctions

R1 disease-causing agent. R2 reduces entry. R3 no: a cell contains many molecules; antibody proteins are made by cells. Teach protein as a type of molecule without requiring amino-acid detail.

## 3 I do Before a specific response

Model: “Mucus traps particles/pathogens. Cilia move mucus towards the throat, helping remove them from the airways.” Contrast with acid’s chemical action. Avoid saying cilia kill pathogens or that all swallowed pathogens die.

## 4 I do Recognise the antigen

Think aloud: “I have a cell that makes proteins, and a protein that binds a target. They are not the same object. The simplified shape lets me predict which target fits; real binding depends on molecular structure.” Antibodies may neutralise/mark; other cells can engulf.

## 5 I do Read the time and amount

Model read x-axis first, then compare y-values at the same elapsed day. "20-2=18 units more, in this constructed dataset. It does not tell me an individual's protection or a universal time." Later exposure is to the same antigen; no universal-antibody claim.

## 6 We do Build the explanation chain

Correct exposure → specific antibody response and formation of memory → memory retained → faster response on later exposure to same antigen. Antibody production and memory formation overlap; avoid a strict claim that memory waits until every antibody disappears.

Reveal: Antigen exposure triggers a specific response and memory formation. Memory retained → faster response to the same antigen later.

## 7 We do Choose a target, test the fit

No: a different antigen may not have a complementary binding site. The drawing models specificity, not actual molecular scale or all immune mechanisms. Let pupils trace/point before revealing.

## 8 We do Evidence desk: compare day 6

6 versus 32 arbitrary units; difference 26. Memory supports faster/larger response to same antigen. Numbers constructed; graph not a clinical forecast.

## 9 Hinge Which mechanism is correct?

B correct. Use protein/cell wordcards if A; refer antigen X/Y if C; distinguish medicine mechanisms if D.

A. Antibodies are memory cells. - Antibodies are proteins; memory lymphocytes are cells.

B. Memory lymphocytes support a faster response to the same antigen. - Yes. Specific memory helps the later response begin sooner and often become larger.

C. One antibody protects equally against every pathogen. - Antibody binding is specific; different antigen features may not be recognised.

D. Antibiotics work by creating immune memory. - Antibiotics act against bacteria. They are not a method for creating specific immune memory.

## 10 You do Create a defence explanation

Set selected route and shared evidence; pupils need no online research.

## 11 You do Make the model teach someone

Supported S1-S4; standard M1-M4; stretch T1-T4. Assess the cell/protein distinction before asking longer explanations. If needed, place lymphocyte and antibody in separate boxes with makes between them.

## 12 Review Audience test: is the mechanism clear?

Allow quiet partner/TA exchange. Pupil improves a label or arrow using the question, or defends an accurate one.

## 13 Review Improve the final claim

Model: vaccination stimulates a specific response and memory; later exposure can be dealt with faster, reducing disease risk/severity, but protection varies and is not guaranteed. Do not introduce schedules or individual advice.

## 14 Exit Explain the faster response

Collect one accurate mechanism each. E 3 should include same antigen and faster response. Revisit matching model if universal protection appears.

## 15 Exit Choose the next useful step

Shared exit minute. Pupils choose privately. No health or vaccination status is requested.

## Answers and assessment

### Retrieval

R1 disease-causing agent. R2 a barrier reduces pathogen entry. R3 cells and proteins are different; proteins are molecules made by cells.

### W1-W3

W1 exposure triggers antibody response and memory formation; memory retained supports a faster later response to the same antigen. W2 no guaranteed binding to Y because its antigen structure differs; shape drawing is a simplified specificity model. W3 day 6 first 6, later 32, difference 26 arbitrary units; memory explanation, constructed-data limit.

### Hinge

A false: protein versus cell. B true: specific memory supports faster secondary response. C false: recognition is specific. D false: antibiotics act against bacterial processes, not by generating specific immune memory.

### S1 / M1

Skin physically reduces entry; mucus traps material and cilia move it towards throat; stomach hydrochloric acid kills many swallowed pathogens; lysozyme in tears can damage bacterial cell walls. M1 requires one physical and one chemical mechanism.

### S2 / M2

Antigen: recognised molecular feature/target that triggers response. Antibody: specific binding protein made by specialised lymphocyte-derived cells. Memory lymphocyte: long-lived cell supporting a faster later response. Core answer need not name B cells/plasma cells.

### S3

Day 4 first 2, later 20;  $20 - 2 = 18$  arbitrary units more in later response.

### S4 / M4

Memory lymphocytes form/are retained; later exposure to the same antigen can trigger faster response. Vaccination supplies antigen or instructions for making antigen, stimulating response and memory without requiring the full disease. Graph is constructed and immunity varies; no individual guarantee.

### M3

Day 6 first 6, later 32; difference 26 arbitrary units. Same-antigen memory supports the faster/larger later response. This is evidence within a teaching model, not clinical proof.

### T1

Specific binding sites need compatible antigen features. Different Y may not bind. Simple shapes omit molecular detail and the many other cell/protein interactions in immunity.

### T2

Later reaches 20 at day 4. First does not reach 20 in the observed 0-10 day interval. Do not say it never reaches 20 or extrapolate beyond the supplied data.

## T3

False inference. Antibody amount can fall while memory lymphocytes persist. Falling proteins are not evidence that every memory cell disappeared.

## T4

Antibiotics act against susceptible bacteria and do not treat viral infection. Vaccination stimulates a specific immune response and memory; it is not an antibiotic. No clinical selection/dosing required.

## Exit and assessment

E 1 accurate barrier+action. E 2 protein. E 3 retained memory lymphocytes support a faster specific response on later exposure to same antigen. L1 accepts a specific correction or evidence-backed confirmation. Secure core requires roles+same antigen+faster response; reteach with cell makes protein if roles are mixed.

## Teaching and assessment notes

40 minutes: opening 2, retrieval 3, I do 10, We do 7, hinge 3, independent 11, review 3, exit 1. Zero-minute slides share their stage time.

Choose one response route. Accept accurate speech, pointing plus explanation, annotations or writing as evidence.

A quiet review still has an audience: teacher or TA reads the pupil's decision, asks one evidence question, and the pupil revises or confirms it.

The antigen/antibody shapes and response curves are explicit conceptual models. Real immune responses involve many cell types and molecules; no universal time, dose or protective antibody threshold is claimed.

Interactive We do: predict an antigen match, test the later-response graph, explain immune memory. Paper equivalent W1-W3 uses the same diagram and data. Animation is learner-controlled and does not model a real person's immune system.

## Access and responsive teaching

### Supported

Same core scientific goal, with word bank, short prompts, read-aloud or pointing/dictation. Do not assign a route from diagnosis.

### Standard

Use the supplied evidence to explain and justify an independent scientific decision.

### Stretch

Optional challenge after secure core: evaluate a limitation and transfer the science. Higher-tier-only material, where present, is labelled explicitly.

### Regulation

Preview the sequence. Offer solo or partner work, a pass and quiet response. Allow a brief reset and re-entry at the next numbered task; no compulsory disclosure, performance or speed competition.

## Misconceptions to check

### **Antibodies are tiny white blood cells.**

Antibodies are proteins; lymphocytes are cells involved in making them.

Check: Ask pupils to put protein or cell beside each word.

### **One antibody fits every antigen.**

Binding is specific to antigen features; the diagram is a simplified fit model.

Check: Offer antigen X and Y in the matching task.

### **Vaccination means infection is impossible.**

Immune memory can give a faster response and reduce disease risk/severity; protection is not an absolute guarantee.

Check: Ask why the graph is not a prediction for every individual.

### **Antibiotics make memory lymphocytes.**

Antibiotics act against bacteria; vaccination stimulates an immune response and memory.

Check: Match each intervention to its mechanism.

| Word              | Meaning                                                                                      |
|-------------------|----------------------------------------------------------------------------------------------|
| barrier           | A defence that reduces entry of pathogens.                                                   |
| antigen           | A molecular feature recognised by the immune system that can trigger a specific response.    |
| antibody          | A protein that binds to a specific antigen and helps defend against it.                      |
| lymphocyte        | A type of white blood cell involved in specific immune responses.                            |
| memory lymphocyte | A long-lived lymphocyte that supports a faster response to a previously encountered antigen. |
| immunisation      | Stimulating protection against a disease, commonly by vaccination.                           |

## Scientific references

### [Pearson Edexcel GCSE Biology 1 BIO specification](#)

Used to check relevant Foundation curriculum content; this is teacher-authored practice, not an official assessment or complete unit. Checked 8 September 2026.

### [WHO: How do vaccines work?](#)

Checked antigen specificity, immune memory and vaccination. Graph values are original constructed numbers, not vaccine-performance data. Checked 8 September 2026.

### [NIAID: Immune system overview](#)

Checked distinction between cells, proteins and innate/adaptive immune functions. Checked 8 September 2026.

### [NHS: Antibiotics](#)

Checked bacteria-versus-virus contrast; no treatment decisions or dosing advice. Checked 8 September 2026.